COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

Set E

Time: 3 Hours **Total Marks: 100**

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form "prove", "show", or "derive" require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. For $x, y \in \mathbb{R}^d$ and $p \geq 1$, recall $d_p(x, y) = \left(\sum_{j=1}^{d} |x_j - y_j|^p\right)^{1/p}$, the distance metric used by the plant's sensor-fusion anomaly detector.

(a) State Minkowski's inequality for ℓ_p norms: $\|u + v\|_p \leq \|u\|_p + \|v\|_p$ for $p \geq 1$.

(b) Using Minkowski's inequality with $u = x - z$ and $v = z - y$ for an arbitrary third point z, prove rigorously that d_p satisfies the triangle inequality $d_p(x, y) \leq d_p(x, z) + d_p(z, y)$ for every finite $p \geq 1$.

(c) Explain precisely why your proof in (b) breaks down for $0 < p < 1$ (identify exactly which step requires $p \geq 1$), and give a brief indication (no full proof required) consistent with the fact that the ℓ_p "distance" for $p < 1$ is not a norm at all.

[8 marks]

Q2. Sensor temperature readings (°C) from $n = 9$ consecutive cycles of one machine are:

$$68,\ 70,\ 71,\ 72,\ 73,\ 74,\ 75,\ 77,\ 150.$$

Compute Q_1, the median, Q_3, and IQR via the median-of-halves convention, and hence the Tukey fences. State which observation(s) are flagged. **[4 marks]**

Q3. A plant's quality-control database suffers a logging bug: a fraction $\varepsilon \in (0, 1)$ of parts are recorded with an identical, coordinated placeholder defect-measurement M (e.g. a sensor timeout code), while the remaining parts have genuine measurements confined to a bounded range negligible relative to M as $M \to \infty$.

(a) Derive $\mu(M)$, $\sigma(M)$, and show that the ordinary z-score of a placeholder measurement converges to $\sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \to \infty$.

(b) Solve for the critical fraction ε^* at which this limit equals 3, and state the operational conclusion for a QC engineer: what fraction of a batch being corrupted identically renders the classical 3σ rule permanently unable to flag it, however extreme the placeholder value M is?

(c) Prove that the modified z-score based on median/MAD instead diverges to $+\infty$ as $M \to \infty$ for any fixed $\varepsilon < 0.5$. State precisely, and distinguish clearly, the two different senses of "breakdown point" at play (single contaminating point with $n \to \infty$ versus a fixed contamination fraction ε with $M \to \infty$), and show that they give numerically different conclusions for this batch.

[9 marks]

Q4. Two machines are described by (Vibration in mm/s, Power Draw in watts): $P = (2.1, 4500)$, $Q = (3.8, 4700)$.

(a) Compute the raw Euclidean distance and identify the dominating coordinate.

(b) Given fleet statistics $\mu_{\text{vib}} = 3.0, \sigma_{\text{vib}} = 0.6$ and $\mu_{\text{pow}} = 4600, \sigma_{\text{pow}} = 80$, standardize both coordinates and recompute the Euclidean distance.

(c) A predictive-maintenance k-NN model is run on raw features. Explain, quoting your two distances, why power draw would completely dominate every neighbour search regardless of the diagnostically important vibration signal.

[6 marks]

Q5. Two machines' logged defect-tags over {Crack, Dent, Discoloration, Warp, Corrosion} are $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

(a) Compute the Jaccard similarity and cosine similarity of x and y.

(b) A third machine's tag vector $z = (0, 0, 0, 0, 0)$ reflects an inspection form that was never filled in, not a genuine claim of zero defects. State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why silently treating z as genuine "no defects" would corrupt a defect-similarity-based root-cause clustering tool.

[4 marks]

Q6. Let R be the missingness indicator for a sensor column $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

(a) State Rubin's formal definitions of MCAR, MAR, and MNAR.

(b) Prove the ignorability result under MAR (with distinct parameters), showing the observed-data likelihood factors cleanly.

(c) Sensors experiencing the most extreme vibration immediately before a bearing failure disproportionately fail to transmit their final reading before going offline, so $P(R = 0 \mid \text{Vibration})$ increases with the true (unrecorded) Vibration value itself. Classify this mechanism formally, and use your proof in (b)–(c) to explain concretely why no imputation using only observed sensor columns can, even in principle, remove the resulting downward bias in average recorded pre-failure vibration.

(d) Suppose the plant also has an independent, always-on low-frequency backup logger correlated with true vibration. Explain, in general terms, how conditioning on this variable could move the analysis toward a recoverable MAR-type situation relative to the enlarged observed set, and why this shows the classification is relative to the observed variable set, not intrinsic to Vibration itself.

[8 marks]

Q7. An analyst applies iterative predictive imputation to five machine-cycle records over Temperature (°C), Pressure (kPa), and Defect Rate (%):

Row	Temperature	Pressure	DefectRate
1	82	101	1.2
2	88	98	**?**
3	79	103	0.9
4	**?**	110	2.1
5	91	97	1.4

Missing cells are initialized with the column mean over observed values. The cyclic order is **DefectRate** $\to$ **Temperature**, using the fitted relations

$\widehat{\text{DefectRate}} = 0.02{\cdot}\text{Temperature}{+}0.01{\cdot}\text{Pressure}{-}1, \qquad \widehat{\text{Temperature}} = 3{\cdot}\text{DefectRate}{-}0.5{\cdot}\text{Pressure}{+}90.$

(a) State the two initial fill values.

(b) Perform one full iteration (Row 2's DefectRate, then Row 4's Temperature), showing all substituted numbers.

(c) Explain, referencing the specific predictor columns involved, why a second iteration leaves both values unchanged here.

(d) Your computed Temperature for Row 4 is markedly lower than every other observed Temperature in the table. Explain what this reveals about the risk of extrapolating a linear predictive-imputation model outside the joint range of predictor values on which it was effectively calibrated, and state one concrete diagnostic check a practitioner should run before trusting such an imputed value.

[6 marks]

Q8. An IoT-platform identity-resolution system, linking device records reported by two independent monitoring platforms, computes pairwise similarities for three device records A, B, C: $s(A, B) = 0.75$, $s(B, C) = 0.73$, $s(A, C) = 0.18$, with lower threshold 0.4 and upper threshold 0.68.

(a) Classify each pair.

(b) State the connected-components clustering of $\{A, B, C\}$ under "match" edges, and explain the transitivity problem.

(c) With $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}}(s(i, j) - \theta)$, $\theta = 0.5$, compute Obj for $\{A, B, C\}$ versus $\{A, B\}, \{C\}$, and conclude which is objectively preferable.

(d) For a chain of n device records with adjacent similarity s_0 (just above upper threshold) and non-adjacent similarity floor f (below lower threshold), derive Obj exactly for (i) the single connected-component clustering and (ii) the clustering retaining only adjacent pairs, and prove the gap is $\Theta(n^2)$. State the operational lesson for a plant merging device identities across many independently deployed monitoring gateways.

[10 marks]

Q9. Repair costs (in Rs.) for nine machine failures are strongly right-skewed: 5, 9, 14, 20, 30, 55, 120, 300, 1
Compute $\log_{10}$ of each to two decimal places, and state the raw ratio max / min against the transformed range, quantifying the compression achieved. **[4 marks]**

Q10. A plant analytics team fits a scaler for *Machine Age (months)* and a one-hot encoder for *Plant Location* on the entire dataset (training and test rows combined) before splitting, to predict machine failure.

(a) With $n_{\text{tr}}, n_{\text{te}}$ the train/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population Machine-Age means restricted to each subpopulation (which may differ if the test set is drawn from a newly commissioned plant), derive the exact bias of the leaked full-data mean relative to the correct train-only mean, and state the condition under which it vanishes.

(b) Explain formally why this bias implies any downstream test-set failure-prediction metric is optimistically biased.

(c) A newly commissioned *Plant Location* appears only among test-time machines. State precisely how a leak-free encoder fitted on training locations only must represent this row, and explain why fitting the encoder on the union of all locations seen across the whole dataset is still leakage.

[8 marks]

Q11. Two candidate duplicate device records are compared on four fields: serial-tag similarity $= 0.95$ (Jaro–Winkler), location agreement $= 1$, ID-tag agreement $=$`NaN` (missing on one monitoring platform), install-date agreement $= 0$. Weights: serial-tag 0.4, ID-tag 0.3, location 0.2, install-date 0.1.

(a) Compute the weighted-mean score using only available fields, weights renormalized.

(b) Classify against thresholds lower $= 0.5$, upper $= 0.75$.

(c) Explain why substituting 0 for the missing ID-tag comparison instead of excluding and renormalizing would bias the decision, and in which direction.

[6 marks]

Q12. Cycle-time readings (seconds) for $n = 7$ production runs: 45, 46, 44, 47, 45, 46, 300.

(a) Compute the mean, standard deviation, and ordinary z-score of the value 300.

(b) Compute the median, MAD, and modified z-score of 300.

(c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 300, and connect the disagreement explicitly to your general result in Q3.

[4 marks]

Q13. A transform $T : D \to \mathbb{R}$, $D \subseteq \mathbb{R}$, is said to *preserve order* on D if $x < y \implies T(x) < T(y)$ for all $x, y \in D$ (i.e. T is strictly increasing on D).

(a) Prove the general lemma: if T is differentiable on an interval D and $T'(x) > 0$ for all $x \in D$, then T is strictly increasing on D (state clearly which classical theorem you invoke).

(b) Using part (a), prove that $T(x) = \log_{10}(x)$ preserves order on $D = (0, \infty)$, and that Min–Max scaling $T(x) = (x - x_{\min})/(x_{\max} - x_{\min})$ and z-score standardization $T(x) = (x - \mu)/\sigma$ both preserve order on any domain where they are defined (i.e. $x_{\max} > x_{\min}$, respectively $\sigma > 0$).

(c) For the Box–Cox family $T_\lambda(x) = (x^\lambda - 1)/\lambda$ ($\lambda \neq 0$, $x > 0$), compute $T'_\lambda(x)$ and prove carefully that it is positive for *every* real $\lambda \neq 0$ and every $x > 0$ (including negative λ), and hence that T_λ preserves order on $(0, \infty)$ regardless of the sign of λ.

(d) Explain why "does one-hot encoding preserve order?" is not merely false but *category-mismatched* as a question (there is no natural total order on the codomain of indicator vectors unless the categories are themselves ordinal), and then prove that any discretization/binning map (which sends an entire sub-interval of D to a single bin label) can **never** be strictly increasing on D whenever that sub-interval contains more than one point of D, while still preserving the induced *weak* (non-strict) order between distinct bins.

[8 marks]

Q14. Two blocking passes generate candidate duplicate device pairs among $n = 8000$ standardized records linking two IoT monitoring platforms. Pass 1 (same plant zone and installation year) yields blocks of sizes $55, 75, 38$. Pass 2 (same last-four hardware-serial digits) yields blocks of sizes $11, 9, 17, 7, 14$. The two candidate sets overlap in 70 pairs.

(a) Compute the candidate pairs from each pass and the size of their union.

(b) Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.

(c) Ground truth contains 350 true duplicate pairs; Pass 1 alone captures 300, Pass 2 alone captures 200, both jointly capture 160. Verify via inclusion–exclusion the pair completeness of the union and state its numerical value.

(d) Prove $1-(1-p_1)(1-p_2) \geq \max(p_1, p_2)$ for independent capture probabilities $p_1, p_2 \in [0, 1]$, and relate it to your answer in (c).

[6 marks]

Q15. A plant reliability engineer gives you eight machine-monitoring records:

ID	Op. Hours	Vibration (mm/s)	Temp (°C)	Defect Rate (%)	Serial Tag
1	1200	2.10	68.0	1.1	line-a-m07
2	1100	2.00	66.0	1.0	line-a-m07
3	3400	2.30	70.0	?	line-b-m12
4	4200	2.60	74.0	2.4	line-c-m03
5	1500	6.80	69.0	1.3	line-a-m15
6	3900	2.40	71.0	?	line-b-m30
7	3500	2.35	70.5	1.9	line-b-m12
8	1300	2.15	68.5	1.2	line-a-m20

Among the six rows with observed Defect Rate, the mean Operating Hours is 2133.3; among the two with missing Defect Rate, the mean Operating Hours is 3650. From the other seven rows (excluding Row 5), $\mu = (\mu_V, \mu_T) = (2.29, 69.71)$ for (Vibration, Temperature) and $\Sigma = \begin{bmatrix} 0.045 & 0.09 \\ 0.09 & 6.5 \end{bmatrix}$. Rows 3 and 7 are suspected duplicate device records logged twice across the two monitoring platforms, with comparison vector: serial-tag similarity 0.95, ID-tag agreement `NaN`, location agreement 1, install-date agreement 0, using the weights/thresholds of Q11.

(a) State, with the standard caveat about MNAR being unprovable from observed data alone, what the operating-hours-conditioned evidence does and does not establish about the missingness mechanism of Defect Rate.

(b) Compute, by hand, the Mahalanobis distance of Row 5's (Vibration, Temperature) $= (6.80, 69.0)$ from μ, explicitly computing Σ^{-1}, and comment on whether Row 5 is a multivariate outlier despite its Temperature looking entirely unremarkable in isolation.

(c) Decide, via the Q11 machinery, whether Rows 3 and 7 should be merged, and construct the merged record's Defect Rate field using the course's numeric golden-record rule.

(d) Argue formally whether deduplication should precede or follow imputation here, prove your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]